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SECTION 1

TUTORIAL APACHE SPARK

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Table of Contents

1.0 Business Understanding.....	3
2.0 Data Understanding (include Relational data model).....	3
3.0 Data Preparation.....	5
4.0 Modeling.....	6
4.1 ETL Architecture Design.....	7
1. Extraction Phase.....	7
2. Transformation Phase.....	7
3. Loading Phase.....	7
4.2 Multidimensional Data Model.....	8
4.3 Fact Table.....	8
Measures.....	9
4.4 Dimension Tables.....	9
4.4.1 Geographic Dimension.....	9
4.4.2 Educational Infrastructure Dimensions.....	10
4.5 Star Schema Design.....	10
4.6 Hierarchies in the Multidimensional Model.....	11
4.6.1 Geographic Hierarchy.....	12
4.6.2 Administrative Hierarchy.....	12
4.6.3 Infrastructure Hierarchy.....	12
4.7 Justification of the Selected Model.....	12
5.0 Evaluation.....	13
6.0 Reflection.....	16
6.1 ANIS SAFIYYA BINTI JANAI.....	16
6.2 NURUL ASYIKIN BINTI KHAIRUL ANUAR.....	16
6.3 LUBNA AL HAANI BINTI RADZUAN.....	16

1.0 Business Understanding

This tutorial was carried out by following the tutorial [Creating a Simple ETL Pipeline With Apache Spark](#) to learn how an ETL pipeline works with Apache Spark. The tutorial uses the Censo Escolar dataset from Brazil, which includes educational data from 2010 to 2021. The dataset includes information about student enrollment, the number of teachers, school facilities and other details.

The main issue identified in this project is the data that is being used was stored in many CSV files and the raw data structure is unsuitable for direct analysis. Many values are repeated and several types of information are combined into a single huge dataset. This makes the data difficult to manage, query, and visualize efficiently.

To address this issue, an ETL pipeline was created. The acronym ETL stands for extract, transform and load. Apache Spark is used to read raw CSV files throughout the extraction process. The transformation procedure organizes the data into a structured format by dividing it into fact and dimension tables. During the loading procedure, the transformed data is stored in PostgreSQL.

Apache Spark was chosen because it processes large datasets more effectively and allows for distributed data processing. Spark also makes it easier to handle numerous CSV files at the same time. PostgreSQL was used to save the transformed data in a relational database format.

In addition, the project applies the star schema concept in data warehousing. The fact table contains measurable data such as total students and teachers, while the dimension tables contain descriptive information such as location, internet access, and school facilities. This structure reduces redundant data and boosts query performance.

After the ETL procedure is completed, the transformed data may be imported into Metabase for visualization and dashboard building. Users can now quickly evaluate trends and generate reports. Overall, the goal of this project is to learn how Apache Spark can be used to create a simple ETL pipeline that converts raw data into a structured format suited for analytics and business intelligence.

2.0 Data Understanding (include Relational data model)

The dataset used in this project is the Brazilian National Basic Education Census dataset collected from 2010 until 2021. The data was obtained from the Brazilian Government Open Data portal in CSV format.

```
print("Total columns:", len(data_csv.columns))
# print("Total rows:", data_csv.count())
```

✓ 0.0s

Total columns: 370

Figure 1: Total columns of the dataset

```
data_csv.show(10)
```

✓ 3.7s

NU_ANO_CENSO	NO_REGIAO	CO_REGIAO	NO_UF	SG_UF	CO_UF	NO_MUNICIPIO	CO_MUNICIPIO	NO_MESORREGIAO	CO_MESORREGIAO	NO_MICRORREGIAO	CO_MICRORREGIAO	CO_DIST
2010	Norte	1	Rondônia	RO	11	Alta Floresta D'O...	1100015	Leste Rondoniense	1102	Cacoal	11006	1106
2010	Norte	1	Rondônia	RO	11	Alta Floresta D'O...	1100015	Leste Rondoniense	1102	Cacoal	11006	1106
2010	Norte	1	Rondônia	RO	11	Alta Floresta D'O...	1100015	Leste Rondoniense	1102	Cacoal	11006	1106
2010	Norte	1	Rondônia	RO	11	Alta Floresta D'O...	1100015	Leste Rondoniense	1102	Cacoal	11006	1106
2010	Norte	1	Rondônia	RO	11	Alta Floresta D'O...	1100015	Leste Rondoniense	1102	Cacoal	11006	1106
2010	Norte	1	Rondônia	RO	11	Alta Floresta D'O...	1100015	Leste Rondoniense	1102	Cacoal	11006	1106
2010	Norte	1	Rondônia	RO	11	Alta Floresta D'O...	1100015	Leste Rondoniense	1102	Cacoal	11006	1106
2010	Norte	1	Rondônia	RO	11	Alta Floresta D'O...	1100015	Leste Rondoniense	1102	Cacoal	11006	1106
2010	Norte	1	Rondônia	RO	11	Alta Floresta D'O...	1100015	Leste Rondoniense	1102	Cacoal	11006	1106
2010	Norte	1	Rondônia	RO	11	Alta Floresta D'O...	1100015	Leste Rondoniense	1102	Cacoal	11006	1106

only showing top 10 rows

Figure 2: 10 rows sample data from the dataset

The dataset consist of information related to schools, enrollments, geographical locations, infrastructure, ethnicity, and education categories. Each CSV file contains approximately 370 columns and around 180MB of yearly data.

After the dataset was explored, the total size of all the datasets combined was around ~2.2GB of raw data which means the datasets are considered semi-large size, making it unsuitable for in-memory processing tools like Pandas.

```
data_csv.printSchema()
```

✓ 0.2s

```
root
|-- NU_ANO_CENSO: integer (nullable = true)
|-- NO_REGIAO: string (nullable = true)
|-- CO_REGIAO: integer (nullable = true)
|-- NO_UF: string (nullable = true)
|-- SG_UF: string (nullable = true)
|-- CO_UF: integer (nullable = true)
|-- NO_MUNICIPIO: string (nullable = true)
|-- CO_MUNICIPIO: integer (nullable = true)
|-- NO_MESORREGIAO: string (nullable = true)
|-- CO_MESORREGIAO: integer (nullable = true)
|-- NO_MICRORREGIAO: string (nullable = true)
|-- CO_MICRORREGIAO: integer (nullable = true)
|-- CO_DISTrito: integer (nullable = true)
|-- CO_ENTIDADE: integer (nullable = true)
|-- NO_ENTIDADE: string (nullable = true)
|-- TP_DEPENDENCIA: integer (nullable = true)
|-- TP_CATEGORIA_ESCOLA_PRIVADA: integer (nullable = true)
|-- TP_LOCALIZACAO: integer (nullable = true)
|-- TP_LOCALIZACAO_DIFERENCIADA: integer (nullable = true)
|-- DS_ENDERECO: string (nullable = true)
|-- NU_ENDERECO: string (nullable = true)
|-- DS_COMPLEMENTO: string (nullable = true)
|-- NO_BAIRRO: string (nullable = true)
|-- CO_CEP: integer (nullable = true)
...
|-- QT_TUR_ESP: integer (nullable = true)
|-- QT_TUR_ESP_CC: integer (nullable = true)
|-- QT_TUR_ESP_CE: integer (nullable = true)
```

Figure 3: The raw dataset schema

Looking at the result of `.printschema()`, the dataset does not follow a fully normalized relational model in its raw form. However, it may contain underlying relational patterns that can be modeled into a star schema for analytical purposes.

The dataset can be conceptually separated into fact and dimension tables. The fact table represents student enrollment metrics and measurement, while dimension tables describe description attributes such as school, geographic location, and time.

Based on the data understanding phase, the dataset was determined to be suitable for a star schema dimensional modelling, allowing effective and efficient analytical querying and dashboard visualization.

3.0 Data Preparation

The data preparation phase involves several processes to ensure the dataset is suitable for analytical processing, dashboard visualization, and decision-making purposes. In this tutorial, the process started by extracting the annual CSV files from the Brazilian Government Open Data portal using Python scripts in batches.

```
urllib3.disable_warnings(urllib3.exceptions.InsecureRequestWarning)

YEARS = [str(i) for i in range(2010, 2022)]

BASE_URL = "https://download.inep.gov.br/dados_abertos/microdados_censo_escolar_(year).zip"

DATA_DIR = "data"
os.makedirs(DATA_DIR, exist_ok=True)

for year in YEARS:
    url = BASE_URL.format(year=year)

    zip_path = os.path.join(DATA_DIR, f"{year}.zip")
    extract_path = os.path.join(DATA_DIR, f"temp_{year}")

    print(f"\nDownloading {year}...")

    # Download ZIP
    response = requests.get(url, stream=True, verify=False)

    with open(zip_path, "wb") as f:
        for chunk in response.iter_content(chunk_size=8192):
            if chunk:
                f.write(chunk)

    print(f"Extracting {year}...")
```

Figure 4: The code snippet for extracting files using URL

To prepare the Spark environment to allow processing data using Spark, Spark application session is created and configured before any files reading, DataFrames creating, and SQL queries running using `SparkSession.builder()`.

```
from pyspark.sql import SparkSession

spark = SparkSession.builder\
    .appName("CensoEscolarStarSchema")\
    .config("spark.sql.shuffle.partitions", "16")\
    .getOrCreate()
```

Figure 5: The Spark session builder code

Next, the raw CSV files were loaded into Apache Spark for processing. Since CSV files are slower and larger for analytical processing, the datasets were converted into Parquet format to improve storage efficiency and query performance.



```
# Read CSV data
data_csv = (
    spark
    .read
    .format("csv")
    .option("header", "true")
    .option("inferSchema", "true")
    .option("delimiter", ";")
    .option("encoding", "iso-8859-1")
    .load("./data/*.csv")
)
```

4] ✓ 28.1s

Figure 6: The code for loading raw data into Apache Spark



```
data_csv.write.mode("overwrite").parquet("./data/censo_escolar.parquet")
```

5] ✓ 1m 35.4s

Figure 7: Format conversion code from CSV file to Parquet files

The data was then transformed into a star schema structure by connecting the fact table with dimension tables using foreign keys. Finally, the prepared data was loaded into PostgreSQL to support dashboard visualization and analytical queries in Metabase.

4.0 Modeling

Using Apache Spark, the modelling phase emphasised on creating an effective ETL architecture and a structured multidimensional data model for educational data analytics. The aim of this stage was to convert raw educational data into an ideal analytical structure that facilitates reporting, institutional evaluation, and multidimensional analysis using the Brazilian school census dataset.

The project used a star schema data warehouse model to ensure scalability and analytical efficiency which adhere to the best practices of data warehousing by distinctly separating quantifiable educational records from descriptive contextual attributes. This method promotes maintainability for future data expansion, streamlines reporting procedures, and improves query performance.

4.1 ETL Architecture Design

Apache Spark was used to develop the ETL pipeline considering its scalability, distributed processing capabilities, and suitability for managing large datasets. It also supported high-performance analytical processing, which facilitates the effective extraction, transformation, and loading of structured data.

There are three main phases for the implemented ETL architecture

1. Extraction Phase

Spark libraries were used in extracting the raw educational census data, which was initially saved in CSV format. School location, internet access, sanitary facilities, educational infrastructure, geographical variables, and other school details were all included in the dataset.

2. Transformation Phase

Spark SQL were used for significant data preprocessing throughout the transformation stage. Numerous methods of data preparation were implemented, such as:

- Remove duplicate item
- Managing inconsistent and missing values
- Standardizing data formats and attribute names
- Removing faulty observations
- Transforming categorical attributes into analytical dimensions
- Creating surrogate keys for dimensional modeling

In order to facilitate multidimensional analysis, the transformation process also normalized descriptive educational data into distinct dimension tables.

3. Loading Phase

The transformed data are then loaded into a star schema structure that was made up from a primary fact table and several dimension tables. This loading process are able to support upcoming business intelligence operations and also increases the effectiveness of analytical queries.

As Apache Spark offers scalable processing, parallel computing, and effective handling of large-scale datasets, therefore, the chosen ETL architecture is ideal for this project. This architecture also indirectly reflects modern big data engineering techniques that are frequently used in data warehouse systems.

4.2 Multidimensional Data Model

A multidimensional star schema model was designed to facilitate analytical reporting and educational insights. This structure was specifically chosen as it reduces query complexity, enhances aggregate efficiency, and facilitates effective Online Analytical Processing (OLAP).

The star schema designed consists of one central fact table, *fact_censo_escolar*, which is linked to other dimension tables that describe geographic features and educational infrastructure.

4.3 Fact Table

The *fact_censo_escolar* table acts as the central fact table and stores measurable educational census data as shown in Table 1.

Table 1: Fact Table Attributes and Description

Attribute	Description
qt_doc_bas	Number of basic education teachers
qt_doc_inf	Number of infant education teachers
qt_doc_fund	Number of elementary education teachers
qt_doc_med	Number of secondary education teachers
qt_mat_bas	Total basic education enrollments
qt_mat_inf	Total infant education enrollments
qt_mat_fund	Total elementary education enrollments
qt_mat_med	Total secondary education enrollments
qt_mat_branca	White student enrollments
qt_mat_preta	Black student enrollments
qt_mat_parda	Mixed-race student enrollments
qt_mat_arela	Asian student enrollments
qt_mat_indigena	Indigenous student enrollments
nu_ano_censo	Census year

The fact table in Table 1 also contains foreign keys referencing all dimension tables.

Measures

The measurable attributes in the fact table support educational performance and demographic analysis. Key measures include:

- Student enrollment counts
- Teacher counts
- Ethnic demographic distributions
- Educational level statistics

These measures enable aggregation operations such as totals, averages, and trend analysis across different dimensions.

4.4 Dimension Tables

The dimensional model includes several descriptive dimension tables representing school infrastructure, educational resources, and geographic information.

4.4.1 Geographic Dimension

This dimension stores geographic information related to educational institutions as described in *Table 2*.

Table 2: Geographic Dimension Table

Attribute	Description
NO_UF	State name
SG_UF	State abbreviation
CO_UF	State code
NO_MUNICIPIO	Municipality name
CO_MUNICIPIO	Municipality code

This dimension enables regional and municipal educational analysis.

4.4.2 Educational Infrastructure Dimensions

Table 3 shows several infrastructure-related dimensions were created to normalize categorical educational attributes.

Table 3: Education Infrastructure Dimension Table

Dimension	Attribute	Description
dim_tp_localizacao	TP_LOCALIZACAO	Stores school location classification either urban or rural.
dim_tp_dependencia	TP_DEPENDENCIA	Represents school administrative dependency which includes federal, state, municipal, or private administration
dim_in_agua_potavel	IN_AGUA_POTAVEL	Represents potable water availability
dim_in_energia_inexistente	IN_ENERGIA_INEXISTENTE	Represents electricity availability.
dim_in_esgoto_inexistente	IN_ESGOTO_INEXISTENTE	Represents sewage system availability.
dim_in_refeitorio	IN_REFEITORIO	Presence of cafeteria facilities.
dim_in_biblioteca	IN_BIBLIOTECA	Presence of library
dim_in_banheiro	IN_BANHEIRO	Presence of bathroom facilities
dim_in_equip_nenhum	IN_EQUIP_NENHUM	Represents absence of educational equipment.
dim_in_internet	IN_INTERNET	Represents internet accessibility.
dim_in_computador	IN_COMPUTADOR	Represents computer availability.

4.5 Star Schema Design

The implemented multidimensional model follows a classic star schema architecture in which the *fact_censo_escolar* table is positioned at the center and connected directly to all dimension tables through surrogate keys.

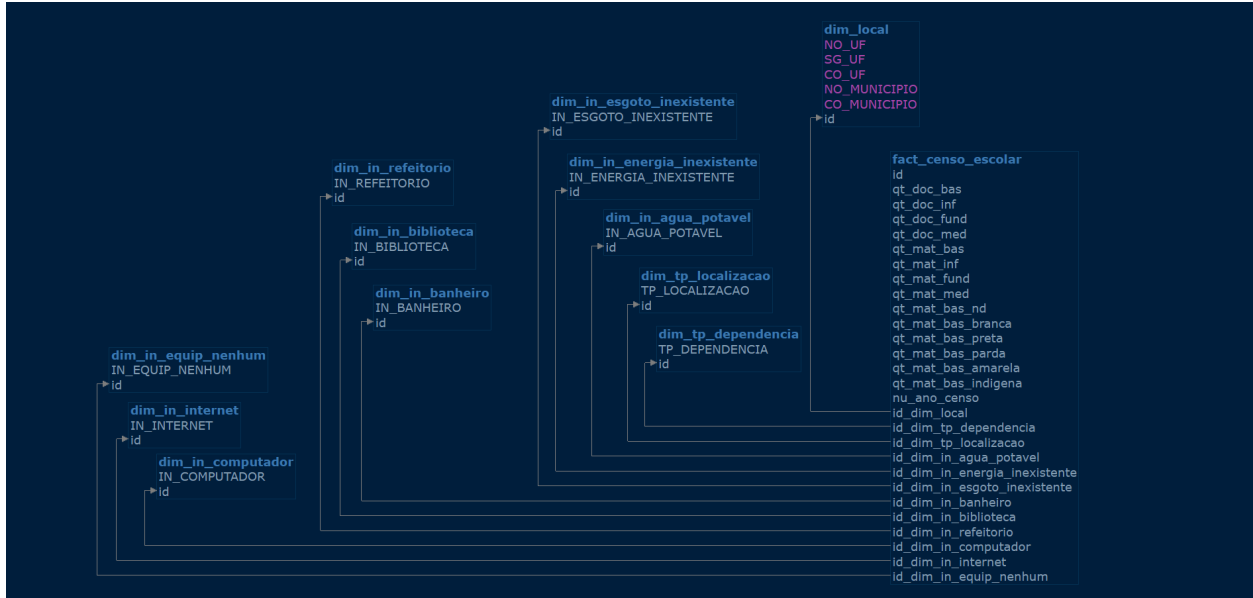


Figure 8: Final data model using Star-Schema

This modeling approach provides several advantages:

- Simplified analytical querying
- Faster aggregation performance
- Reduced join complexity
- Improved scalability
- Enhanced maintainability
- Efficient OLAP operations

The use of separate dimension tables for infrastructure attributes improves normalization and allows educational indicators to be analyzed independently across multiple perspectives.

The schema design also reflects dimensional modeling best practices because measurable educational data is clearly separated from descriptive contextual attributes.

4.6 Hierarchies in the Multidimensional Model

Hierarchies were incorporated into the dimensional structure to support drill-down and roll-up analytical operations.

4.6.1 Geographic Hierarchy

State → Municipality

This hierarchy enables educational analysis at both the state and municipal levels

4.6.2 Administrative Hierarchy

Administrative Dependency → School Location

This hierarchy supports comparative analysis between public and private institutions across urban and rural regions.

4.6.3 Infrastructure Hierarchy

Infrastructure Availability → Educational Resource Type

This hierarchy enables evaluation of school facilities and technological readiness.

The inclusion of hierarchies significantly improves multidimensional analytical capability and supports advanced reporting operations.

4.7 Justification of the Selected Model

The selected architecture and multi-dimensional model are highly suitable for the needs of the project, because they meet the established data warehouse principles and educational analysis goals. Using Apache Spark can improve scalability, processing performance and conversion efficiency, while the star model structure enhances the performance of analytical queries and reports.

In addition, the realized multi-dimensional model shows the powerful dimensional modeling practice in the following ways:

- Clearly distinguish between fact tables and dimensional tables
- Use meaningful educational indicators
- Realize the logical hierarchy
- Support OLAP analysis
- Reduce redundancy through dimensional separation
- Improve scalability and maintainability
- Realize efficient educational reporting and analysis

The design also supports future scalability, allowing the integration of more educational indicators and dimensions without major structural modifications.

In short, the modeling method successfully converts the original education census data into a structured analysis framework, which can support institutional analysis, infrastructure evaluation, demographic reporting and large-scale educational decision-making.

5.0 Evaluation

The ETL pipeline was successfully executed using Apache Spark and PostgreSQL. The dataset from 2010 until 2021 was loaded from multiple CSV files and processed using the Spark library. The The ETL pipeline was successfully executed with Apache Spark and PostgreSQL. Several CSV files containing educational datasets from 2011 to 2021 were extracted and the Spark library was used for processing. The modified data was then put into PostgreSQL as fact and dimension tables.

During the execution process, all dimension tables, including dim_local, dim_tp_dependencia and other infrastructure-related dimensions were successfully connected with the fact table via foreign keys. The final fact table, fact_censo_escolar, was constructed correctly and without significant problems. The pipeline could also process huge educational datasets efficiently with Apache Spark.

In order to assess the ETL pipeline's performance, runtime testing was done using Apache Spark to compare CSV and Parquet file formats. The goal of this test was to determine which file format performs best for large data processing.

Figure 9 shows the runtime results of processing the dataset in CSV format. The recorded processing time was about 11.79 seconds.

```
from time import time

start = time()

csv_data = (
    spark
    .read
    .format("csv")
    .option("header", "true")
    .option("inferSchema", "true")
    .load("./data/*.csv")
)

csv_data.count()

end = time()

print("CSV Runtime:", end - start, "seconds")
✓ 11.7s
CSV Runtime: 11.785073041915894 seconds
```

Figure 9: CSV Runtime Result

Figure 10 shows the runtime results of processing the dataset in Parquet format. The recorded processing time was about 1.72 seconds.

```
start = time()

parquet_data = (
    spark
    .read
    .format("parquet")
    .load("./data/censo_escolar.parquet/")
)

parquet_data.count()

end = time()

print("Parquet Runtime:", end - start, "seconds")
✓ 1.7s
Parquet Runtime: 1.7166476249694824 seconds
```

Figure 10: Parquet Runtime Result

The comparison showed that the Parquet format performed much faster than the CSV format. This is because Parquet stores data in a columnar format that is ideal for analytical processing in

Apache Spark. Parquet requires less storage than CSV and allows Spark to read only the relevant columns rather than scanning the full file.

In addition, the structured data saved in the PostgreSQL was linked with the dashboard in the Metadata for visualization. The dashboard used charts and graphs to help present educational information clearly. For example, the Figure 11 shows a dashboard representation of total student enrollment by year. The visualization makes it easy to track enrollment trends from 2010 to 2021.

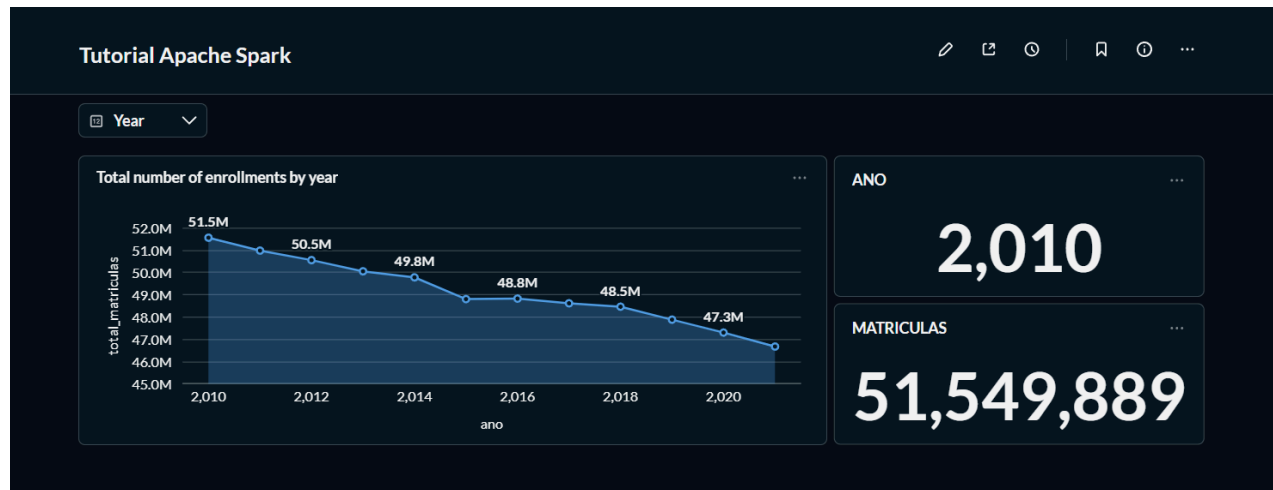


Figure 11: Sum Enrollment by Year Dashboard

However, there are several limitations to this tutorial. The ETL pipeline was only tested on a local machine environment, therefore runtime performance may differ between machines depending on hardware factors such as CPU performance, RAM capacity and storage speed. Furthermore, the tutorial focused on batch processing and did not cover real-time streaming data processing.

Overall, the ETL pipeline effectively achieved its goal of extracting, processing and loading the educational data into a structured data warehouse format using Apache Spark. The evaluation results also showed that Apache Spark and Parquet format can improve the processing performance for large-scale data analysis.

6.0 Reflection

This section demonstrates the individual reflection for this tutorial.

6.1 ANIS SAFIYYA BINTI JANAI

Completing this tutorial provides an in-depth understanding of Apache Spark to implement scalable data pipelines. By converting the Censo Escolar data set from a fragmented, row-oriented CSV file to a structured star model, data redundancy can be minimized and also optimize data architecture, so as to support more refined analytical queries. This tutorial also shows the technical knowledge of data cleaning and relational modeling in the big data environment. The significant performance difference between different storage types during pipeline execution is also an important discovery in this learning process. By switching the processing layer from the original CSV format to the Parquet format, the computing runtime is optimized from 11.79 seconds to 1.72 seconds, which shows how columnar storage and data compression can significantly reduce input and output overhead. The project also shows how structured ETL pipelines can directly improve business intelligence and decision-making, bridging the gap between theoretical data warehouses and practical applications.

6.2 NURUL ASYIKIN BINTI KHAIRUL ANUAR

This tutorial provided a better understanding of how a data pipeline works with Apache Spark. It demonstrated on how the ETL is used in actual data processing using large datasets, where previously it was only understood in theory. This was the first exposure to Apache Spark in Python and using commands like `spark.read()`. Some issues appeared early on when working with the Parquet format, but they were fixed after troubleshooting. Another problem was ensuring that various CSV files were properly combined into fact and dimension tables. Overall, this project enhanced comprehension of data engineering concepts, particularly those related to data processing, organizing, and analysis.

6.3 LUBNA AL HAANI BINTI RADZUAN

This tutorial provided exposure to Apache Spark and simple ETL pipeline development. Several challenges were encountered during the process, especially in relation to Windows environment setup and storage management. The tutorial also provides exposure to new visualization tools such as Metabase and data warehouses such as Adminer using PostgreSQL. This tutorial enhances the technical understanding of data stream processing, ETL processes and visualization tools. In addition, experience was gained in transforming raw datasets into optimized structures using Parquet format and relational database loading. Overall, the project strengthened both technical and analytical skills in data engineering and business intelligence workflows. Future improvements may include implementing automated workflows, cloud-based storage, and more advanced dashboard analytics.